

CTF CHALLENGE

BLUE{report}

CHALLENGE ANSWERS & REPORT

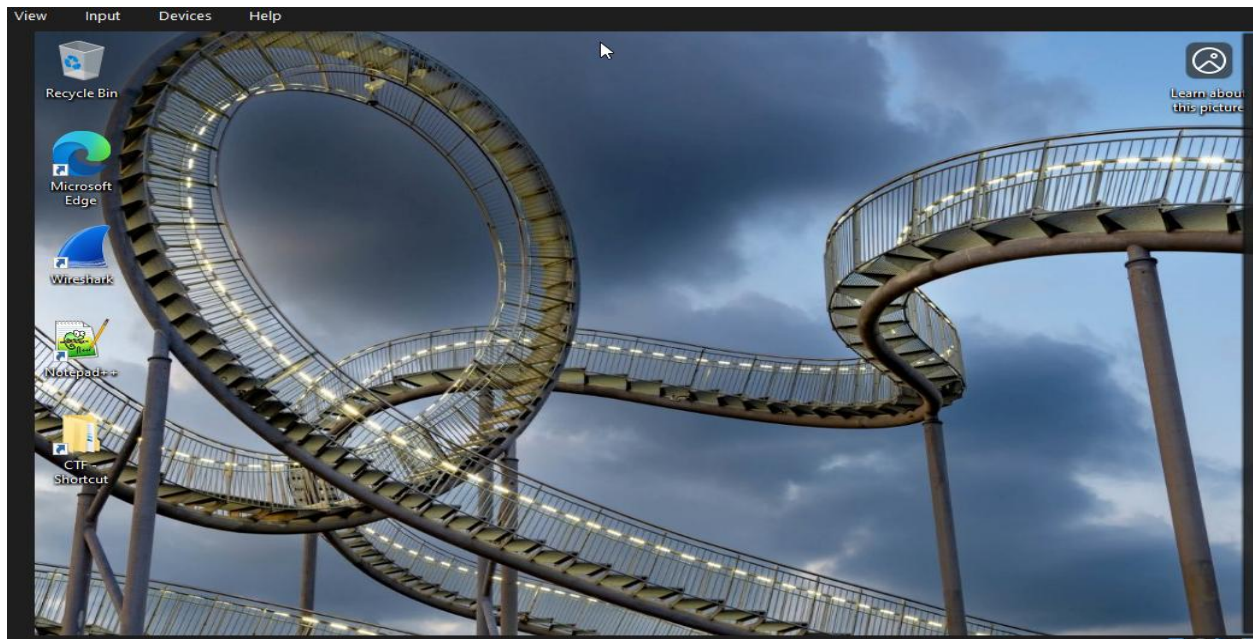
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Snode CTF Ideathon 2026

Blue Team Operations

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The Running Environment



In this overall task, we were using the following tools:

VMware, Wireshark, Notepad++, and Event-Viewer.

Report/Approach to the Challenge

1. Analyzing logs and datasets

I start by going through the logs from top to bottom to understand what kind of data I'm working with. I look for key event types (such as logins, account creation, or network connections) and filter or search for relevant entries using tools like Notepad++ or Wireshark. I also group similar events together so it's easier to compare normal vs abnormal activity.

2. Identifying suspicious patterns or anomalies

Once I understand the normal activity, I look for anything that stands out. This could be unusual login times, repeated connections at fixed intervals, unexpected processes, or activity from unknown IP addresses. I also compare similar entries like multiple svchost.exe processes to spot differences that might indicate something suspicious.

3. Detecting and validating threats

When I find something unusual, I investigate it further by tracing it across different data sources. For example, I might follow a suspicious account through logs or check a process across multiple Volatility outputs. I validate the threat by confirming multiple indicators (e.g., strange behavior, unusual network connection, suspicious memory activity) instead of relying on just one sign.

4. Prioritizing and responding to incidents

I prioritize issues based on how risky they are. Anything involving admin access, external connections, or possible data exposure is treated as high priority. Once identified, I would recommend actions such as isolating the affected system, removing malicious accounts or processes, and documenting all findings clearly so the team can respond quickly and prevent further impact.

Challenge 1- Suspicious Authentication Event

Q1- What Event ID represents a successful windows logon?

4624

Q2- What time did the suspicious logon occur?

2026/03/30 10:03:51

Q3- What logon Type number was used, and what does it mean?

Logon Type 10- It usually means that a remote desktop may have made a connection, and it can be a malicious log. It is a very suspicious activity because an attacker might be using stolen credentials via an RDP.

Q4- What was the source IP address of the suspicious logon?

192.168.10.55

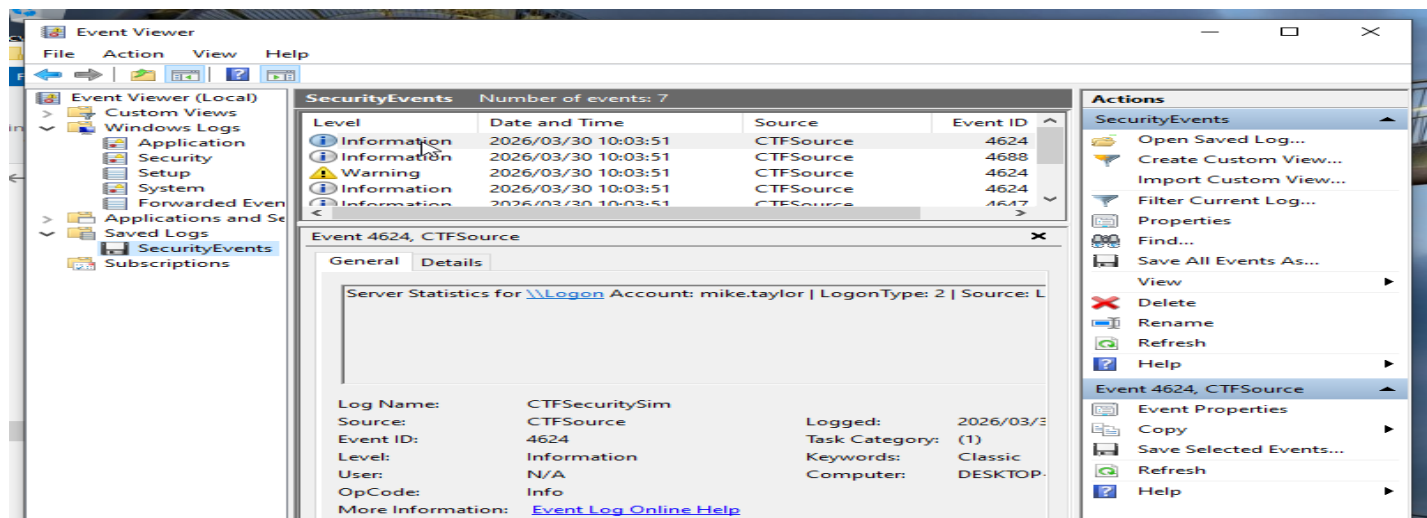
Q5- What field inside the suspicious logon contains the flag?

Data field

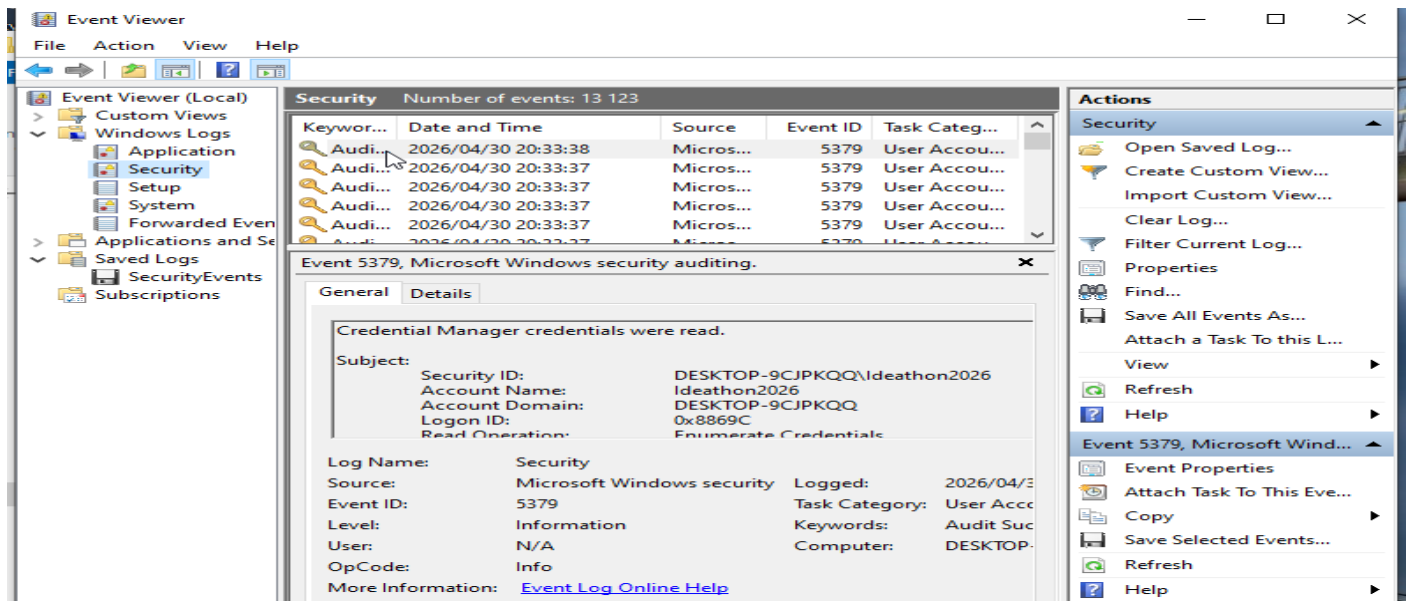
Q6- What was the flag?

CTF{afterhours_rdp_login_detected}

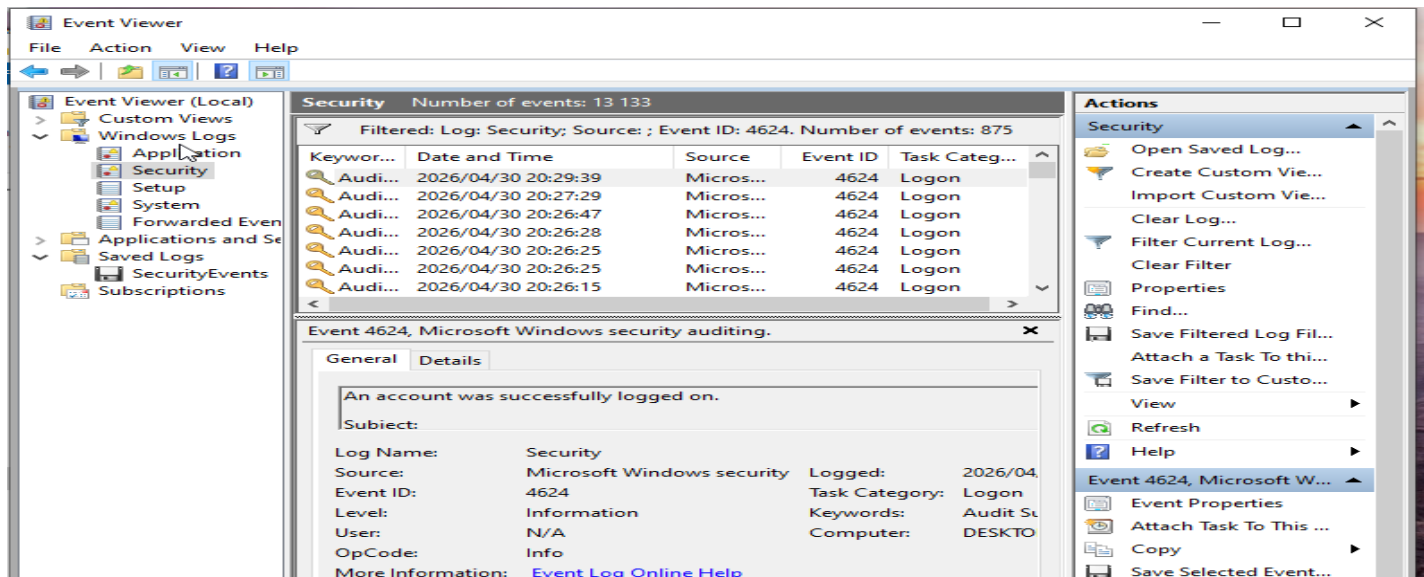
This is the environment of Challenge 1- Event Viewer



The events and logs unfiltered



Event IDs filtered to the event ID 4624- which is a logon



Challenge 2 – Anomalous Outbound Communication Pattern

Q1- Which internal IP address is generating suspicious traffic?

Ip Address 10.0.0.45

Q2- What is the destination IP address of the suspicious connections?

Ip Address 10.0.0.45

Q3- What is the exact time interval in seconds between each beacon packet?

60seconds .

Q4- What HTTP method is the beacon using, and what path does it send to?

POST/ update, Path- 1.1

Q5- What field in the HTTP body contains the flag?

Check_in=true&flag=CTF{c2_beacon_60s_interval} answer = Flag

This is the working environment for Challenge 2

The image shows a Wireshark packet capture of a beacon. The packet list shows a POST request to /api from 10.0.0.23 to 172.217.0.1. The packet details pane shows the HTTP body containing the flag: Check_in=true&flag=CTF{c2_beacon_60s_interval}.

No.	Time	Source	Destination	Protocol	Leng	Info
1	0.000000	10.0.0.23	172.217.0.1	HTTP	115	GET / HTTP/1.1
2	30.000000	10.0.0.23	172.217.0.1	HTTP	128	GET /search?q=news HTTP/1.1
3	30.000000	10.0.0.10	10.0.0.1	HTTP	127	POST /api HTTP/1.1
4	30.000000	10.0.0.67	52.96.88.1	HTTP	123	GET /owa HTTP/1.1
5	30.000000	10.0.0.45	185.220.101.33	HTTP	182	POST /update HTTP/1.1
6	60.000000	10.0.0.45	185.220.101.33	HTTP	182	POST /update HTTP/1.1
7	60.000000	10.0.0.45	185.220.101.33	HTTP	182	POST /update HTTP/1.1
8	60.000000	10.0.0.45	185.220.101.33	HTTP	205	POST /update HTTP/1.1
9	60.000000	10.0.0.45	185.220.101.33	HTTP	182	POST /update HTTP/1.1
10	60.000000	10.0.0.45	185.220.101.33	HTTP	182	POST /update HTTP/1.1
11	100.000000	10.0.0.12	13.107.42.14	HTTP	118	GET / HTTP/1.1
12	50.000000	10.0.0.23	172.217.0.1	HTTP	118	GET /mail HTTP/1.1

Frame 1: Packet, 115 bytes on wire (920 bits), 115 bytes captured
Ethernet II, Src: VMware_11:22:33 (00:0c:29:11:22:33), Dst: VMware
Internet Protocol Version 4, Src: 10.0.0.23, Dst: 172.217.0.1
Transmission Control Protocol, Src Port: 52100, Dst Port: 80, Seq
Hypertext Transfer Protocol

Challenge 3 – Malicious Scheduled Task detected

Q1- What are the full names of both tasks in the file?

<Task name="Microsoft\Windows\WindowsUpdate\ScheduledStart">

<Task name="Microsoft\Windows\WindowsUpdate\AutoUpdateCheck">

Q2- What is the command path the suspicious task runs, and why is it unusual?

<Command>C:\Users\Public\winupd.exe</Command>, It is unusual because it is located in a user folder.

Q3- Who is listed as the author of the suspicious task?

<Author>Corp\Lisa.Turner</Author>

Q4- What date and time was the suspicious task created?

Date: 2024-02-05 Time: 09:30:00

Q5- which field contains the flag?

The source field,

<Source>CTF{persist_schtask_found_T1053}</Source>

This is the environment for challenge 3

```
C:\CTF\Challenge3\scheduled_tasks_export.xml - Notepad++
file Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
new 1 scheduled_tasks_export.xml account_audit.log install.log
1 <?xml version="1.0" encoding="UTF-16"?>
2 <Tasks>
3
4 <Task name="\Microsoft\Windows\WindowsUpdate\ScheduledStart">
5 <RegistrationInfo>
6 <Author>Microsoft Corporation</Author>
7 <Description>Starts the Windows Update service when needed.</Description>
8 <URI>\Microsoft\Windows\WindowsUpdate\ScheduledStart</URI>
9 </RegistrationInfo>
10 <Triggers>
11 <BootTrigger>
12 <Enabled>true</Enabled>
13 </BootTrigger>
14 </Triggers>
15 <Actions>
16 <Exec>
17 <Command>%SystemRoot%\system32\sc.exe</Command>
18 <Arguments>start wuauserv</Arguments>
19 </Exec>
20 </Actions>
21 <Settings>
22 <Enabled>true</Enabled>
23 <RunOnlyIfIdle>false</RunOnlyIfIdle>
24 </Settings>
25 </Task>
26
27 <Task name="\Microsoft\Windows\WindowsUpdate\AutoUpdateCheck">
28 <RegistrationInfo>
29 <Author>CORP\lisa.turner</Author>
30 <Description>Checks for available updates and applies them silently.</Description>
31 <URI>\Microsoft\Windows\WindowsUpdate\AutoUpdateCheck</URI>
```

Challenge 4- Rogue Admin Account Detected

Q1- How many accounts were created during this 24-Hour window?

3 accounts

Q2- Which account was created without a ServiceNow change ticket?

Haxor

Q3- What group was the rogue account immediately added to?

Administrators

Q4- What eventID later in the log is associated with the rogue account and what does it represent?

ID=4624, It represents a Logon. The account is haxor.

Q5- What encoding is used on the RegValue field and what does it decode to?

From base64 and it decodes to CTF{rogue_admin_discovered}

This is the environment for challenge 4

```
C:\CTF\Challenge4\account_audit.log - Notepad++
file Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
new 1 scheduled_tasks_export.xml account_audit.log install.log
1 int=john.smith LogonType=2 Source=LOCAL WorkstationName=CORP-WS01
2 int=sarah.jones LogonType=2 Source=LOCAL WorkstationName=CORP-WS04
3 int=helpdesk.temp Created_By=IT_ADMIN@corp.local ServiceNow_Ticket=CHG0012445
4 int=helpdesk.temp Group=HelpDesk Added_By=IT_ADMIN@corp.local
5 int=mike.taylor LogonType=2 Source=LOCAL WorkstationName=CORP-WS11
6 int=peter.walsh Reason=BadPassword Caller=CORP-DC-01
7 int=jenny.admin LogonType=10 Source=192.168.1.55 WorkstationName=UNKNOWN
8 int=haxor Created_By=jenny.admin@corp.local ServiceNow_Ticket=NONE
9 int=haxor Group=Administrators Added_By=jenny.admin@corp.local
10 int=haxor PasswordExpiry=Never LastModified=jenny.admin@corp.local
11 int=jenny.admin Image=cmd.exe CommandLine=whoami
12 int=jenny.admin Image=cmd.exe ExitCode=0
13 int=haxor FailureCode=0x18 CallerIP=192.168.1.55
14 int=haxor LogonType=3 Source=192.168.1.55 WorkstationName=UNKNOWN
15 int=haxor CallerProcess=net.exe GroupName=Administrators
16 intName=HKLM\SYSTEM\CurrentControlSet\Services RegValue=C1RGe3UvZ3VlX2xvY2FeX2FkbWluX2Rpc2NvdnVvZWR5 ModifiedBy=haxor
17 int=contractor.2024 Created_By=IT_ADMIN@corp.local ServiceNow_Ticket=CHG0012501
18 int=contractor.2024 Group=HelpDesk Added_By=IT_ADMIN@corp.local
19 int=john.smith LogonType=2 Source=LOCAL WorkstationName=CORP-WS01
20 int=sarah.jones LogoffTime=11:02:33 WorkstationName=CORP-WS04
21
```

Challenge 5- Memory Dump Analysis- Volatility 3 Detected

Q1- Which PID stands out in the pslist section and why?

PID 1337 with 1 thread, it stands out because all the other svchost.exe processes have multiple threads.

Q2- what is missing from that PID's command line that all other svchost entries have?

The k-service group parameter is missing.

Q3- What external IP and port is that PID connected to?

Ip: 185.220.101.33 Port 4444

Q4- What memory protection flag on that PID indicates code injection

Protection: PAGE_EXECUTE_READWRITE

Q5- What is the encoded string in the extracted strings, and what does it decode to?

String : Q1RGe3Byb2N1c3NfaG9sbG93aW5nX3N2Y2hvc3RfY2F1Z2h0fQ==

From base64, Decodes to CTF{proccuss_hollowing_svchost_caught}

Environment for challenge 5

```
volatility_output - Notepad
File Edit Format View Help
-----
Volatility 3 - Memory Analysis Report
Image: EXEC-PC-01_memdump.raw
OS: Windows 10 x64 Build 19041
Captured: 2024-01-12 08:05:33 UTC
-----
I
[+] Running windows.pslist
-----
PID      PPID     Name           Threads  Wow64
4         0        System        139      False
524       4        smss.exe      3        False
632       524      csrss.exe     12       False
704       524      wininit.exe   3        False
716       632      winlogon.exe  5        False
876       704      services.exe  9        False
884       704      lsass.exe     7        False
492       876      svchost.exe   18       False
628       876      svchost.exe   14       False
856       876      svchost.exe   22       False
1044      876      svchost.exe   11       False
1132      876      svchost.exe   9        False
1337      704      svchost.exe   1        False
1508      876      spoolsv.exe   7        False
1604      876      svchost.exe   13       False
1888      876      svchost.exe   8        False
-----
Ln 1, Col 1    100%    Unix (LF)    UTF-8
```

